

Infinite-failure Software Reliability Models Based on Non-homogeneous Markov Processes

Nanxiang Qiu

m222093@hiroshima-u.ac.jp

Hiroshima University

Siqiao Li

Hiroshima University

Tadashi Dohi

Hiroshima University

Hiroyuki Okamura

Hiroshima University

Research Article

Keywords: software reliability model, infinite-failure model, non-homogeneous Markov processes, software intensity function, goodness-of-fit performance, predictive performance

Posted Date: November 1st, 2023

DOI: <https://doi.org/10.21203/rs.3.rs-3507541/v1>

License: © ⓘ This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at Software Quality Journal on September 17th, 2026. See the published version at <https://doi.org/10.1007/s11219-026-09772-5>.

Infinite-failure Software Reliability Models Based on Non-homogeneous Markov Processes

Nanxiang Qiu^{1*}, Siqiao Li^{1†}, Tadashi Dohi^{1†}, Hiroyuki Okamura^{1†}

^{1*}Graduate School of Advanced Sciences and Engineering, Hiroshima University, 1-4-1 Kagamiyama, Higashi-Hiroshima, 739-8527, Hiroshima, Japan.

*Corresponding author(s). E-mail(s): m222093@hiroshima-u.ac.jp;
Contributing authors: loyaltotribe@gmail.com; dohi@hiroshima-u.ac.jp;
okamu@hiroshima-u.ac.jp;

[†]These authors contributed equally to this work.

Abstract

This paper explores yet another software reliability modeling framework based on non-homogeneous Markov processes (NHMPs). For two subclasses of NHMPs; generalized binomial processes (GBPs) and generalized Polya processes (GPPs), we formulate 22 novel NHMP-based software reliability models (SRMs) with 11 kinds of baseline intensity functions, which are different from the existing NHMP-based SRMs by Li et al. (2023). Our evaluation of NHMP-based SRMs focuses on assessing their goodness-of-fit and predictive performances using 8 data sets of software fault detection time-domain data and 8 data sets of time-interval data (group data). The results are compared with the well-known NHPP-based SRMs. Through comprehensive numerical experiments, we show that our new modeling framework could provide the strengths in the goodness-of-fit performance in many cases but outperform on the predictive performance in a limited number of cases.

Keywords: software reliability model; infinite-failure model; non-homogeneous Markov processes; software intensity function; goodness-of-fit performance; predictive performance

1 Introduction

Software reliability modeling is a critical technique for estimating quantitative software reliability, which is defined as the probability of a software system operating without failure over a specific period of time. It involves stochastic modeling and statistical inference to quantify the software reliability as well as the fault count behavior of software systems in both testing and operational phases. Also, software reliability modeling helps software developers and engineers to understand the testing progress based on the cumulative number of software faults detected in the testing. Since the reliable functionality of software is essential for ensuring the whole system dependability and the efficient operation of various systems, the quantitative assessment of software reliability plays a central role in quality assurance, especially, in mission-critical systems. Over the last five decades, a large number of software reliability models (SRMs) have been developed and widely used to describe the cumulative number of software faults detected during the testing phase [1–3].

Non-homogeneous Poisson process (NHPP)-based SRMs [4–6, 8–10, 15, 17–19, 21, 22, 29, 30] are commonly used to describe the occurrence of software failures caused by software faults over the testing time, under the assumption that the software intensity function varies with time. The NHPP-based SRMs can also be classified into two types; finite-failure SRMs and infinite-failure SRMs [1–3]. More specifically, if the mean value function denoting the expected cumulative number of software faults detected in the testing is bounded, i.e., the number of inherent faults remaining in the software before the testing is given by a Poisson distributed random variable with finite mean, then the corresponding NHPP-based SRMs are called the finite-failure NHPP-based SRMs. On one hand, if a software failure caused by a software fault is minimally repaired by fixing the fault, and the software system just after fixing the fault is restored back to a working condition that is only as good as it was just before the software failure, then the number of software faults through the life cycle monotonically increases and the mean value function denoting the expected cumulative number of software faults detected in the testing is unbounded. Such models are called the infinite-failure NHPP-based SRMs. The logarithmic Poisson execution time SRM by Musa and Okumoto [17] is representative of this modeling category. Li et al. [13] compared the infinite-failure NHPP-based SRMs with finite-failure ones [22] with same 11 time to software-failure distribution functions in a comprehensive numerical experiment with actual software fault count data and showed the superiority of NHPP-based SRMs over the existing NHPP-based SRMs. In fact, the NHPP-based SRMs are rather specific stochastic models. For instance, it is assumed that the transition rates from one state to another state in NHPPs depend on only time. This implies that the software intensity function varies over time even when no detection/fixing of faults occurs. From another point of view in statistics, the expected cumulative number of software faults in the NHPP-based SRMs are exactly equivalent to their variance. Since the uncertainty on the number of software faults decreases as the testing time goes on, the above specific property is questionable for describing the software fault count processes.

Very recently, Li et al. [14] generalized the NHPP-based SRMs by introducing non-homogeneous Markov processes (NHMPs) and developed a unified modeling framework to describe the software fault count phenomena. More concretely, for two

subclasses of NHMPs; generalized binomial processes (GBPs) [25, 26] and generalized Polya processes (GPPs) [12], they formulated 14 GBP-based SRMs and 14 GPP-based SRMs, and compared them with their associated finite-failure NHPP-based SRMs. In this way, the NHMP-based SRMs by Li et al. [14] are considered generalizations of the finite-failure NHPP-based SRMs. These models are proposed by assuming that the cumulative intensity functions are bounded on time, so every software debugging is perfect, which means that the precondition of using these models is that no new faults will be generated during debugging. In the NHPP-based modeling, the mean value function of the NHPP-based SRM is finite. However, imperfect debugging occurs frequently in real situations, so we need to think of another model to deal with this situation. In the similar line with the logarithmic Poisson execution time NHPP-based SRM by Musa and Okumoto [17], an extension of the infinite-failure models should be investigated as well. The same authors [24] tried to examine other NHMP-based SRMs with unbounded cumulative intensity functions in a limited number of data sets and competitive SRMs.

This paper is a continuation of our conference paper [24] and aims at evaluating the model performance in a comprehensive experiment. More specifically, we propose another software reliability modeling framework based on NHMPs including GBPs and GPPs, which can be categorized into the infinite-failure models. In our modeling, we show that while our new GBP-based SRMs generalized the existing infinite-failure NHPP-based SRMs, they do not possess the same characteristics as the infinite-failure model. It is worth mentioning that our NHMP-based SRMs in this paper are quite different from the existing NHMP-based SRMs [14], because the state transition rates in our NHMP-based SRMs are derived in a fashion similar to the infinite-failure NHPP-based SRMs. We formulate respective 11 GBP- and GPP-based SRMs with 11 kinds of baseline intensity functions, and assess the goodness-of-fit and predictive performances of our NHMP-based SRMs using 8 data sets of software fault detection time-domain data and 8 data sets of software fault detection time-interval data (group data). In other words, the NHMP-based SRMs by Li et al. [14] are generalizations of the finite-failure NHPP-based SRMs [22], but our NHMP-based SRMs in this paper are generalizations of infinite-failure NHPP-based SRMs [13].

The remaining part of this paper is organized as follows. Section 2 summarizes the well-known NHPP-based SRMs from the viewpoint of finite- and infinite failure models. In Section 3 we introduce the NHMP-based SRMs by Li et al. [14] and propose the different versions of NHMP-based SRMs with the same software-failure time distributions as [14]. Section 4 is devoted to a numerical experiment with actual software fault count data, where the results are compared with the infinite-failure NHPP-based SRMs [13] and other existing finite-failure SRMs. Throughout the comparative study, we show that our new modeling framework could provide the strengths in the goodness-of-fit performance in many cases but outperform on the predictive performance in a limited number of cases.

2 NHPP-based Software Reliability Modeling

2.1 Preliminary

The non-homogeneous Poisson process (NHPP) is one of the simplest stochastic counting processes used to describe the occurrence of rare events over time. In the context of software reliability, these rare events correspond to software-failure occurrences caused by software-fault detection. It is assumed that the occurrence of software failures follows a Poisson process, where the number of failures in a fixed time interval is a Poisson distributed random variable. However, unlike a homogeneous Poisson process (HPP), the intensity of an NHPP is not constant and varies with time.

Suppose that the software test starts at time $t = 0$. Let $\{N(t), t \geq 0\}$ be the cumulative number of software faults detected by time t (≥ 0). The stochastic process $N(t)$ is said an NHPP if the following conditions hold.

- $N(0) = 0$,
- $\{N(t), t \geq 0\}$ has independent increment,
- $\Pr\{N(t + \Delta t) - N(t) \geq 2\} = o(\Delta t)$,
- $\Pr\{N(t + \Delta t) - N(t) = 1\} = \lambda(t; \boldsymbol{\theta})\Delta t + o(\Delta t)$,

where the function $\lambda(t; \boldsymbol{\theta})$ represents an absolutely continuous function, known as the intensity function, $\boldsymbol{\theta}$ denotes the model parameter (vector), and $o(\Delta t)$ represents the higher-order term of the infinitesimal time Δt , satisfying $\lim_{\Delta t \rightarrow 0} o(\Delta t)/\Delta t = 0$.

Let $P_n(t) = \Pr\{N(t) = n | N(0) = 0\}$ be the transition probabilities of $N(t)$. From the Markovian property, the Kolmogorov forward equations are obtained by

$$\frac{d}{dt}P_0(t) = \lambda(t; \boldsymbol{\theta})P_0(t), \quad (1)$$

$$\frac{d}{dt}P_n(t) = \lambda(t; \boldsymbol{\theta})P_{n-1}(t) - \lambda(t; \boldsymbol{\theta})P_n(t), \quad n = 1, 2, \dots \quad (2)$$

with the initial conditions $P_0(0) = 1$ and $P_n(0) = 0$ ($n = 1, 2, \dots$). By the inductive argument, we have the probability mass function of $N(t)$ as

$$P_n(t) = \frac{\{\Lambda(t; \boldsymbol{\theta})\}^n}{n!} \exp(-\Lambda(t; \boldsymbol{\theta})), \quad n = 0, 1, 2, \dots \quad (3)$$

where

$$E[N(t)] = \sum_{n=0}^{\infty} nP_n(t) = \Lambda(t; \boldsymbol{\theta}) = \int_0^t \lambda(x; \boldsymbol{\theta})dx. \quad (4)$$

The function $\Lambda(t; \boldsymbol{\theta})$ is referred to as the mean value function, representing the expected cumulative number of software faults found by time t in the NHPP-based SRMs.

2.2 Finite- and Infinite-failure Models

In finite-failure NHPP-based SRMs, the assumption is made such that each fault in a software program is detected at independent and identically distributed random time, following a continuous cumulative distribution function $F(t; \boldsymbol{\pi})$ with parameter $\boldsymbol{\pi}$. Additionally, it is assumed that the initial number of residual software faults in the software system, say at time $t = 0$, is finite. Moreover, we suppose that the initial number of residual software faults before testing obeys a Poisson distribution with mean ω (> 0). From a simple algebraic manipulation, it is easy to see that the cumulative number of software faults $N(t)$ is given by an NHPP with mean value function $\Lambda(t; \boldsymbol{\theta}) = \omega F(t; \boldsymbol{\pi})$ with $\boldsymbol{\theta} = (\omega, \boldsymbol{\pi})$. It is evident to confirm that $\lim_{t \rightarrow \infty} \Lambda(t; \boldsymbol{\theta}) = \omega$, if $\lim_{t \rightarrow \infty} F(t; \boldsymbol{\pi}) = 1$, so the expected cumulative number of software faults even after a sufficiently long testing time is finite. Okamura and Dohi [22] developed SRATS; a Software reliability assessment tool on the spreadsheet, by implementing 11 representative time to software-failure distributions in the finite-failure NHPP-based SRMs.

In the above finite-failure NHPP-based SRMs, it is implicitly assumed that no new software faults are introduced during the software debugging, which limits their applicability to the scenario where each debugging may be imperfect. However, in practice, this assumption may not hold in actual software testing processes. In contrast, the infinite-failure NHPP-based SRMs do not require such a strong assumption. This is because the infinite-failure NHPP-based SRMs assume that the software intensity function increases continuously over time due to the imperfect debugging, and that the occurrence of a software failure does not affect the future behavior of the fault count process. Li et al. [13] investigated the model performances for the infinite-failure NHPP-based SRMs with the same 11 software-failure time distributions in SRATS [22].

Suppose that the intensity function $\lambda(t; \boldsymbol{\theta})$ is given by a failure rate of the time to software fault, $f(t; \boldsymbol{\pi})/\{1 - F(t; \boldsymbol{\pi})\}$, with $\boldsymbol{\theta} = \boldsymbol{\pi}$, where $f(t; \boldsymbol{\pi}) = dF(t; \boldsymbol{\pi})/dt$ is the probability density function. Then, it holds that

$$F(t; \boldsymbol{\pi}) = 1 - \exp\{-\Lambda(t; \boldsymbol{\pi})\}. \quad (5)$$

So the cumulative intensity function in the infinite-failure NHPP-based SRMs is equivalent to the mean value function, and is given by

$$E[N(t)] = \Lambda(t; \boldsymbol{\pi}) = -\ln(1 - F(t; \boldsymbol{\pi})). \quad (6)$$

It can be easily checked that $\lim_{t \rightarrow \infty} \Lambda(t; \boldsymbol{\pi}) \rightarrow \infty$, so the mean value functions of the infinite-failure NHPP-based SRMs are always unbounded. In the common lifetime data analysis, the definition in Eq.(6) is common and the assumption of $\Lambda(t; \boldsymbol{\theta}) = \omega F(t; \boldsymbol{\pi})$ is seldom used. In another word, the bounded mean value functions have been validated in only software reliability engineering. However, as Musa et al. [2] and Li et al. [13] showed, the infinite-failure modeling assumption could be validated even in describing software fault-count processes. This fact motivates us to extend the infinite-failure NHPP-based SRMs from the standpoint of generalized stochastic process modeling.

2.3 Parameter Estimation

The parameters of the intensity function, θ ($\theta = \pi$ for the infinite-failure NHPP-based SRMs), can be estimated via various statistical methods. Especially, the commonly used technique in software reliability engineering is the maximum likelihood estimation, by maximizing the log-likelihood function. Here, we suppose that software fault detection time data or software fault detection time-interval data (group data) are available. Taking $t_0 = 0$ as a precondition, we record the occurrence time of each software fault, capturing up to m time points in each data set. For the time data $(t_1, t_2, \dots, t_m)$, we estimate the model parameters θ in the NHPP-based SRMs using the maximum likelihood estimation, where the expressions for the likelihood function and the log-likelihood function are given by

$$L_{NHPP}(\theta) = \prod_{i=1}^m \lambda(t_i; \theta) \times e^{-\Lambda(t_m; \theta)}, \quad (7)$$

$$\ln L_{NHPP}(\theta) = \sum_{i=1}^m \ln \lambda(t_i; \theta) - \Lambda(t_m; \theta), \quad (8)$$

respectively.

In another scenario, let (t_i, n_i) ($i = 1, 2, \dots, m$) be the group data, where t_i denotes the i -th testing time, n_i is the cumulative number of software faults detected by time t_i , and m is the number of data. Then the likelihood function and the log-likelihood function are given as

$$L_{NHPP}(\theta) = \prod_{i=1}^m \left[\frac{[\Lambda(t_i; \theta) - \Lambda(t_{i-1}; \theta)]^{n_i - n_{i-1}}}{(n_i - n_{i-1})!} \right] \times e^{-[\Lambda(t_i; \theta) - \Lambda(t_{i-1}; \theta)]}, \quad (9)$$

$$\begin{aligned} \ln L_{NHPP}(\theta) = & \sum_{i=1}^m \ln \{ (n_i - n_{i-1})! \ln \{ \Lambda(t_i; \theta) - \Lambda(t_{i-1}; \theta) \} \} - \sum_{i=1}^m \ln \{ (n_i - n_{i-1})! \} \\ & - \lambda(t_m; \theta), \end{aligned} \quad (10)$$

respectively, where $(t_0, n_0) = (0, 0)$. Finally, the maximum likelihood estimate of the model parameter is given by $\hat{\theta} = \arg \max \ln L_{NHPP}(\theta)$.

3 NHMP-based Software Reliability Modeling

3.1 Definition

In the NHPP-based SRMs, the transition rate is given by the intensity function itself which depends on only testing time t . While, the well-known homogeneous Markov processes (HMPs) have the state-dependent transition rates that are independent of time. However, in order to describe real-world complex systems, it is not enough to

assume only the time-varying or the state-dependent behavior. To address this limitation for NHPPs and HMPs, a natural extension of the existing SRMs is to consider the time- and state-dependent transition rates, and to introduce a non-homogeneous Markov process (NHMP). For the related references on NHMP-based SRMs, see [14]. In the NHMP-based SRMs, the transition rates depend on not only time but also the current state of the system, i.e., $\lambda_n(t; \boldsymbol{\theta})$ ($n = 0, 1, 2, \dots$). This generalization framework allows us more flexible and realistic modeling of many dynamical systems, such as communication networks and biological systems. In general, solving the corresponding Kolmogorov forward equations for the NHMP-based SRMs is not trivial, as we may not know the explicit form of the probability mass function $P_n(t)$. Li et al. [14] decomposed the state-dependent and time-dependent terms in the NHMP transition rates, and supposed two linear transition rates (intensity functions) as

$$\lambda_n(t; \boldsymbol{\theta}) = (\beta - \alpha n)\kappa(t; \boldsymbol{\pi}), n = 0, 1, 2, \dots, \gamma_{\alpha, \beta}, \quad (11)$$

$$\lambda_n(t; \boldsymbol{\theta}) = (\beta + \alpha n)\kappa(t; \boldsymbol{\pi}), n = 0, 1, 2, \dots, \quad (12)$$

where $\alpha (\geq 0)$ and $\beta (\geq 0)$ are non-negative real parameters, $\boldsymbol{\theta} = (\alpha, \beta, \boldsymbol{\pi})$, $\boldsymbol{\pi}$ is a parameter vector in $\kappa(t; \boldsymbol{\pi})$ which is an absolutely continuous function of t , $\gamma_{\alpha, \beta} = \lceil \beta/\alpha \rceil$, and $\lceil x \rceil$ is the ceiling function, which denotes the largest integer less than x . We call $\kappa(t; \boldsymbol{\pi})$ and $\Lambda(t; \boldsymbol{\pi}) = \int_0^t \kappa(x; \boldsymbol{\pi}) dx$ the baseline intensity function and the cumulative baseline intensity function, respectively. The above linear NHMPs were discussed by Kendall [11] and Konno [12]. Especially, the NHMP-based SRMs are characterized by transition rates in Eqs. (11) and (12) are known as the generalized binomial processes (GBPs) and the generalized Polya processes (GPPs), respectively.

3.2 GBP-based SRMs

The GBP was first considered as an inverse death-process type of NHMP by Kendall [11]. Shanthikumar [25, 26] applied the GBP to represent the software fault count processes, whose Kolmogorov forward equations are given by

$$\frac{d}{dt}P_0(t) = -\beta\kappa(t; \boldsymbol{\pi})P_0(t), \quad (13)$$

$$\begin{aligned} \frac{d}{dt}P_n(t) &= \{\beta - \alpha(n-1)\}\kappa(t; \boldsymbol{\pi})P_{n-1}(t) \\ &\quad - \{\beta - \alpha n\}\kappa(t; \boldsymbol{\pi})P_n(t), n = 1, 2, \dots, \gamma_{\alpha, \beta}, \end{aligned} \quad (14)$$

$$\frac{d}{dt}P_{\gamma_{\alpha, \beta}}(t) = \{\beta - \alpha(\gamma_{\alpha, \beta} - 1)\}\kappa(t; \boldsymbol{\pi})P_{\gamma_{\alpha, \beta}-1}(t). \quad (15)$$

Solving the differential equations in Eqs. (13)-(15) with the initial conditions $P_0(0) = 1$ and $P_n(0) = 0$, one obtains

$$\begin{aligned} P_n(t) &= \binom{\gamma_{\alpha, \beta}}{n} A(t; \alpha, \boldsymbol{\pi})^{\gamma_{\alpha, \beta} - n} (1 - A(t; \alpha, \boldsymbol{\pi}))^n, \\ n &= 0, 1, \dots, \gamma_{\alpha, \beta}, \end{aligned} \quad (16)$$

where

$$A(t; \alpha, \boldsymbol{\pi}) = e^{-\alpha \Lambda(t; \boldsymbol{\pi})} \quad (17)$$

and

$$\Lambda(t; \boldsymbol{\pi}) = \int_0^t \kappa(x; \boldsymbol{\pi}) dx. \quad (18)$$

It is worth to see that Eq. (16) is a binomial distribution with

$$\mathbb{E}[N(t)] = \gamma_{\alpha, \beta} \{1 - e^{-\alpha \Lambda(t; \boldsymbol{\pi})}\}, \quad (19)$$

$$\text{Var}[N(t)] = \gamma_{\alpha, \beta} \{1 - e^{-\alpha \Lambda(t; \boldsymbol{\pi})}\} e^{-\alpha \Lambda(t; \boldsymbol{\pi})}. \quad (20)$$

We easily confirm that $\mathbb{E}[N(t)] > \text{Var}[N(t)]$, so that GBP has the under-dispersion property with $N(t) = 0, 1, 2, \dots, \gamma_{\alpha, \beta}$. Li et al. [14] proposed the NHMP-based SRMs by assuming that the baseline intensity function $\kappa(t; \boldsymbol{\pi})$ is given by an absolutely continuous probability density function $f(t; \boldsymbol{\pi}) = dF(t; \boldsymbol{\pi})/dt$. Then, one sees $\Lambda(t; \boldsymbol{\pi}) = F(t; \boldsymbol{\pi})$ and $A(t; \alpha, \boldsymbol{\pi}) = \exp\{-\alpha F(t; \boldsymbol{\pi})\}$. Hence, it is confirmed that $0 \leq A(t; \alpha, \boldsymbol{\pi}) \leq \exp\{-\alpha\}$ and that the resulting binomial probability is right-truncated. Further, we have

$$\mathbb{E}[N(t)] = \gamma_{\alpha, \beta} \{1 - e^{-\alpha F(t; \boldsymbol{\pi})}\}, \quad (21)$$

$$\text{Var}[N(t)] = \gamma_{\alpha, \beta} \{1 - e^{-\alpha F(t; \boldsymbol{\pi})}\} e^{-\alpha F(t; \boldsymbol{\pi})}. \quad (22)$$

It can be checked that $\lim_{t \rightarrow \infty} \mathbb{E}[N(t)] = \gamma_{\alpha, \beta} \{1 - \exp(-\alpha)\}$ and $\lim_{t \rightarrow \infty} \text{Var}[N(t)] = \gamma_{\alpha, \beta} \{1 - \exp(-\alpha)\} \exp(-\alpha)$, and that the GBP-based SRMs in [14] contain the existing NHPP-based SRMs with bounded mean value functions [1].

On the other hand, in our new NHMP-based SRMs, let $\kappa(t; \boldsymbol{\pi})$ be the failure rate, i.e., $\kappa(t; \boldsymbol{\pi}) = f(t; \boldsymbol{\pi})/\{1 - F(t; \boldsymbol{\pi})\}$. Then we have

$$\Lambda(t; \boldsymbol{\pi}) = -\ln\{1 - F(t; \boldsymbol{\pi})\}, \quad (23)$$

$$A(t; \alpha, \boldsymbol{\pi}) = \{1 - F(t; \boldsymbol{\pi})\}^\alpha. \quad (24)$$

From the results above, we find that $0 \leq A(t; \alpha, \boldsymbol{\pi}) \leq 1$, and that the mean value function is still bounded, because $\lim_{t \rightarrow \infty} \mathbb{E}[N(t)] = \gamma_{\alpha, \beta}$ and $\lim_{t \rightarrow \infty} \text{Var}[N(t)] = 0$. Hence, our alternative GBP-based SRMs seem to be quite reasonable, because $\text{Var}[N(t)]$ approaches 0 as the testing time goes on. However, even though the alternative GBP-based SRMs are extensions of the infinite-failure NHPP-based SRMs, it does not have the key feature that the mean value functions are unbounded. Therefore, we cannot categorize the above GBP-based SRMs into infinite-failure models, but yet alternative GBP-based SRMs.

3.3 GPP-based SRMs

In contrast to the GBP-based SRMs, the generalized Polya process (GPP) is regarded as a pure birth-process within the framework of NHMP by Konno [12], which may be called the generalized Yule process (see Gat [7]). The associated Kolmogorov forward equations are given by

$$\frac{d}{dt}P_0(t) = -\beta\kappa(t; \boldsymbol{\pi})P_0(t), \quad (25)$$

$$\begin{aligned} \frac{d}{dt}P_n(t) = & \{\beta + \alpha(n-1)\}\kappa(t; \boldsymbol{\pi})P_{n-1}(t) \\ & - \{\beta + \alpha n\}\kappa(t; \boldsymbol{\pi})P_n(t), n = 1, 2, \dots \end{aligned} \quad (26)$$

Solving the differential equations in Eqs. (25) and (26) with the initial conditions $P_0(0) = 1$ and $P_n(0) = 0$, one gets

$$\begin{aligned} P_n(t) = & \binom{n+\gamma_{\alpha,\beta}-1}{n} A(t; \alpha, \boldsymbol{\pi})^{\gamma_{\alpha,\beta}} (1 - A(t; \alpha, \boldsymbol{\pi}))^n, \\ & n = 0, 1, \dots \end{aligned} \quad (27)$$

It is seen that Eq.(27) is equivalent to a negative binomial distribution with

$$E[N(t)] = \gamma_{\alpha,\beta} \{e^{\alpha\Lambda(t; \boldsymbol{\pi})} - 1\}, \quad (28)$$

$$\text{Var}[N(t)] = \gamma_{\alpha,\beta} \{e^{\alpha\Lambda(t; \boldsymbol{\pi})} - 1\} e^{\alpha\Lambda(t; \boldsymbol{\pi})}. \quad (29)$$

We also confirm that $E[N(t)] < \text{Var}[N(t)]$, so that GPP exhibits the over-dispersion, which is a desirable property for modeling uncertainty in time series analysis. By the similar calculation procedure to the GBP-based SRMs, we can find that $\lim_{t \rightarrow \infty} E[N(t)] = \gamma_{\alpha,\beta} \{\exp(\alpha) - 1\}$ and $\lim_{t \rightarrow \infty} \text{Var}[N(t)] = \gamma_{\alpha,\beta} \{\exp(\alpha) - 1\} \exp(\alpha)$, or in another way, $\lim_{t \rightarrow \infty} E[N(t)] = \lim_{t \rightarrow \infty} \text{Var}[N(t)] \rightarrow \infty$. It is noted that the GPP-based SRMs do not have an absorbing state $\gamma_{\alpha,\beta}$. It should be known that the GPP is also an extension of a mixed Poisson process model by Okamura and Dohi [20] as well as the infinite-failure NHPP-based SRMs [13]. It is clear that both GBP and GPP are generalizations of HMPs and NHPPs. In particular, when $\kappa(t; \boldsymbol{\pi}) = 1$ and $\alpha = 0$, HMP and NHPP are equivalent to special cases of GBP and GPP, respectively. It is important to note that the failure rates $\kappa(t; \boldsymbol{\pi})$ of our infinite-failure NHMP-based SRMs are completely different from those of the existing finite-failure NHMP-based SRMs [14]. This is a remarkable difference from the existing SRMs.

We give the cumulative distribution functions and intensity functions of the 14 baseline models used in the finite-failure NHMP-based SRMs [14] in Table I, where 11 finite-failure models and 3 infinite-failure models are involved. In our infinite-failure NHMP-based SRMs, we assume the 11 baseline intensity functions in Table II similar to [13]. As an example, the baseline intensity functions of 11 infinite-failure models are illustrated in Figure 1 with $b = 2$ and $c = 1.5$. It is seen that the four typical behaviors; constant, decreasing, increasing, and unimodal properties with respect to testing time can be described.

Table 1 Representative baseline intensity functions of finite-failure SRMs.

Models	$\Lambda(t; \pi), \pi = (b, c)$
Exponential dist. (exp)	$\Lambda(t; \pi) = \omega F(t; \pi)$ $F(t; \pi) = 1 - e^{-bt}$
Gamma dist. (gamma)	$\Lambda(t; \pi) = \omega F(t; \pi)$ $F(t; \pi) = \int_0^t \frac{c^b s^{b-1} e^{-cs}}{\Gamma(b)} ds$
Pareto dist. (pareto)	$\Lambda(t; \pi) = \omega F(t; \pi)$ $F(t; \pi) = 1 - (\frac{b}{b+t})^c$
Truncated normal dist. (tnorm)	$\Lambda(t; \pi) = \omega \frac{F(t; \pi) - F(0; \pi)}{1 - F(0; \pi)}$ $F(t; \pi) = \frac{1}{\sqrt{2\pi}b} \int_{-\infty}^t e^{-\frac{(s-c)^2}{2b^2}} ds$
Log-normal dist. (lnorm)	$\Lambda(t; \pi) = \omega F(\ln t; \pi)$ $F(t; \pi) = \frac{1}{\sqrt{2\pi}b} \int_{-\infty}^t e^{-\frac{(s-c)^2}{2b^2}} ds$
Truncated logistic dist. (tlogist)	$\Lambda(t; \pi) = \omega \frac{F(t; \pi) - F(0; \pi)}{1 - F(0; \pi)}$ $F(t; \pi) = \frac{1}{1 + e^{-\frac{t-c}{b}}}$
Log logistic dist. (llogist)	$\Lambda(t; \pi) = \omega F(\ln t; \pi)$ $F(t; \pi) = \frac{1}{1 + e^{-\frac{t-c}{b}}}$
Truncated extreme-value max dist. (txvmax)	$\Lambda(t; \pi) = \omega \frac{F(t; \pi) - F(0; \pi)}{1 - F(0; \pi)}$ $F(t; \pi) = e^{-e^{-\frac{t-c}{b}}}$
Log-extreme-value max dist. (lxvmax)	$\Lambda(t; \pi) = \omega F(\ln t; \pi)$ $F(t; \pi) = e^{-e^{-\frac{t-c}{b}}}$
Truncated extreme-value min dist. (txvmin)	$\Lambda(t; \pi) = \omega \frac{F(0; \pi) - F(-t; \pi)}{F(0; \pi)}$ $F(t; \pi) = e^{-e^{-\frac{t-c}{b}}}$
Log-extreme-value min dist. (lxvmin)	$\Lambda(t; \pi) = \omega(1 - F(-\ln t; \pi))$ $F(t; \pi) = e^{-e^{-\frac{t-c}{b}}}$
Logarithmic Poisson (log)	$\Lambda(t; \pi) = \Lambda(t; \pi) = a \ln(1 + bt)$
Power-law (Plaw)	$\Lambda(t; \pi) = at^b$
Generalized truncated logist. dist. (gtlogist)	$\Lambda(t; \pi) = \omega F(t; \pi)$ $F(t; \pi) = \frac{1 - e^{-at^c}}{1 + be^{-at^c}}$

* $\Gamma(\cdot)$: standard gamma function, $\text{erf}(\cdot)$: error function.

3.4 Parameter Estimation

Similar to the parameter estimation of NHPP-based SRMs, we use the maximum likelihood method to estimate the parameters $\theta = (\alpha, \beta, \pi)$ of GBP- and GPP-based SRMs, where $t_0 = 0$ is always assumed as a precondition. For m time data that records the time of occurrence of each fault such as $(t_1, t_2, \dots, t_m)$, we can obtain the likelihood

Table 2 Baseline intensity functions of our infinite-failure NHMP-based SRMs.

Models	$\Lambda(t; \boldsymbol{\pi}), \boldsymbol{\pi} = (b, c)$
Exponential dist. (exp)	bt
Gamma dist. (gamma)	$\ln(\Gamma(b)) - \ln(\Gamma(b, \frac{t}{c}))$
Pareto dist. (pareto)	$-b(\ln(c) - \ln(c+t))$
Truncated normal dist. (tnorm)	$\ln(\operatorname{erf}(\frac{c}{\sqrt{2b}}) + 1) - \ln(\operatorname{erf}(\frac{c-t}{\sqrt{2b}}) + 1)$
Log-normal dist. (lnorm)	$-\ln(\operatorname{erf}(\frac{c-\ln(t)}{\sqrt{2b}}) + 1) + \ln(2)$
Truncated logistic dist. (tlogist)	$\ln(e^{c/b} + e^{t/b}) - \ln(e^{c/b} + 1)$
Log logistic dist. (llogist)	$\ln(e^{c/b} + t^{1/b}) - c/b$
Truncated extreme-value max dist. (txvmax)	$\ln(1 - e^{-e^{c/b}}) - \ln(1 - e^{-e^{\frac{c-t}{b}}})$
Log-extreme-value max dist. (lxvmax)	$-\ln(1 - e^{-e^{\frac{c-\ln(t)}{b}}})$
Truncated extreme-value min dist. (txvmin)	$-\ln(e^{-e^{c/b}(e^{t/b}-1)})$
Log-extreme-value min dist. (lxvmin)	$e^{\frac{c+\ln(t)}{b}}$

* $\Gamma(\cdot)$: standard gamma function, $\operatorname{erf}(\cdot)$: error function.

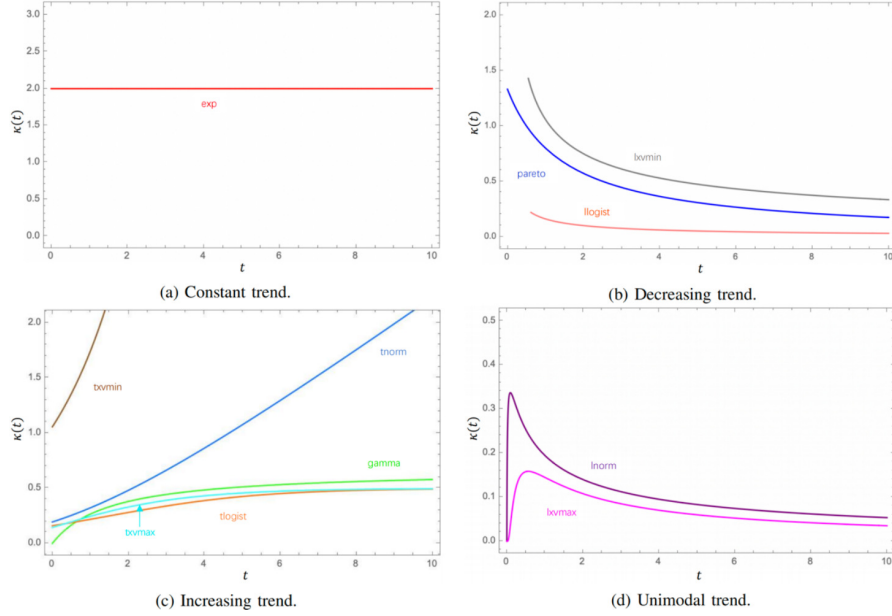


Fig. 1 Some examples of $\kappa(t)$ with $b = 2$ and $c = 1.5$.

function and the log-likelihood function of the GBP from Eq.(11) as

$$L_{GBP}(\boldsymbol{\theta}) = \prod_{i=1}^m \{\beta - \alpha(i-1)\} \kappa(t_i; \boldsymbol{\pi}) \times e^{-\{\beta - \alpha(i-1)\} \{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}, \quad (30)$$

$$\begin{aligned} \ln L_{GBP}(\boldsymbol{\theta}) &= \sum_{i=1}^m \ln\{\beta - \alpha(i-1)\} + \sum_{i=1}^m \ln \kappa(t_i; \boldsymbol{\pi}) \\ &\quad - \sum_{i=1}^m \{\beta - \alpha(i-1)\} \\ &\quad \times \{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}. \end{aligned} \quad (31)$$

On the other hand, consider m group data (t_i, n_i) ($i = 1, 2, \dots, m$), where t_i represents the i -th testing time and n_i denotes the cumulative number of software faults detected by time t_i . Then we can derive the likelihood function and log-likelihood function of GBP from Eq. (16) as

$$L_{GBP}(\boldsymbol{\theta}) = \prod_{i=1}^m \binom{\gamma_{\alpha, \beta} - n_{i-1}}{n_i - n_{i-1}} \times [e^{-\alpha\{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}]^{\gamma_{\alpha, \beta} - n_i} \times [1 - e^{-\alpha\{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}]^{n_i - n_{i-1}}, \quad (32)$$

$$\begin{aligned} \ln L_{GBP}(\boldsymbol{\theta}) &= \sum_{i=1}^m \ln\{(\gamma_{\alpha, \beta} - n_{i-1})! - (n_i - n_{i-1})! \\ &\quad - (\gamma_{\alpha, \beta} - n_i)!\} - \sum_{i=1}^m (\gamma_{\alpha, \beta} - n_i) \times \alpha\{\Lambda(t_i; \boldsymbol{\pi}) \\ &\quad - \Lambda(t_{i-1}; \boldsymbol{\pi})\} + \sum_{i=1}^m (n_i - n_{i-1}) \\ &\quad \times \ln[1 - e^{-\alpha\{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}]. \end{aligned} \quad (33)$$

Next, we consider the GPP. From Eq. (12) we obtain the likelihood function and log-likelihood function of GPP with time data as

$$L_{GPP}(\boldsymbol{\theta}) = \prod_{i=1}^m \{\beta + \alpha(i-1)\} \kappa(t_i; \boldsymbol{\pi}) \times e^{-\{\beta + \alpha(i-1)\} \{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}, \quad (34)$$

$$\begin{aligned}
\ln L_{GPP}(\boldsymbol{\theta}) &= \sum_{i=1}^m \ln\{\beta + \alpha(i-1)\} + \sum_{i=1}^m \ln \kappa(t_i; \boldsymbol{\pi}) \\
&\quad - \sum_{i=1}^m \{\beta + \alpha(i-1)\} \\
&\quad \times \{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}.
\end{aligned} \tag{35}$$

We also get the likelihood function and log-likelihood function of GPP from Eq.(27) when using the group data as

$$\begin{aligned}
L_{GPP}(\boldsymbol{\theta}) &= \prod_{i=1}^m \binom{\gamma_{\alpha,\beta} + n_i - 1}{n_i - n_{i-1}} \\
&\quad \times [e^{-\alpha\{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}]^{\gamma_{\alpha,\beta} + n_{i-1}} \\
&\quad \times [1 - e^{-\alpha\{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}]^{n_i - n_{i-1}},
\end{aligned} \tag{36}$$

$$\begin{aligned}
\ln L_{GPP}(\boldsymbol{\theta}) &= \sum_{i=1}^m \ln\{(\gamma_{\alpha,\beta} + n_i - 1)! - (n_i - n_{i-1})! \\
&\quad - (\gamma_{\alpha,\beta} + n_{i-1} - 1)!\} - \sum_{i=1}^m (\gamma_{\alpha,\beta} + n_{i-1}) \\
&\quad \times \alpha\{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\} + \sum_{i=1}^m (n_i - n_{i-1}) \\
&\quad \times \ln[1 - e^{-\alpha\{\Lambda(t_i; \boldsymbol{\pi}) - \Lambda(t_{i-1}; \boldsymbol{\pi})\}}].
\end{aligned} \tag{37}$$

By maximizing $\ln L_{GPP}(\boldsymbol{\theta})$ and $\ln L_{GPP}(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$, we obtain the maximum likelihood estimates $\hat{\boldsymbol{\theta}}$ in respective cases.

4 Performance Comparison

In order to validate the effectiveness of our novel GBP- and GPP-based SRMs in Eqs. (16) and (27), we conduct numerical experiments to compare them with the existing NHPP-based SRMs [13] and finite-failure GBP/GPP-based SRMs [13]. In the preliminary results [24], we could not perform the comprehensive comparison of all the finite- and infinite- failure SRMs in both NHPP- and NHMP-based modeling. Here, we add the actual data sets in the empirical comparison and explore the superiority of non-homogeneous Markovian modeling.

4.1 Data Sets

We analyze 8 software fault-detection time-domain data (time data)(TDS1~TDS8) in Table 3, and 8 software fault-detection time-interval data (group data) (GDS1~GDS8) in Table 4, which are measured in actual software development projects. Specifically, TDS1, TDS2, TDS6, GDS1, and GDS2 are observed from real-time command and control systems. TDS3 is observed from military application, and a failure in such a

system has the potential to have serious repercussions. The faults recorded in TDS4, TDS7, and TDS8 originate from the single-user workstation. GDS3 and GDS4 are observed in the system testing of tandem software systems. The detected faults in GDS5 are measured in an inertial navigating system with a navigation technology capable of independently measuring, calculating, and tracking the position, direction, and velocity of a moving object. GDS6~GDS8 are measured in the testing of the embedded applications for the printer.

Table 3 Software fault-detection time data sets.

Data	No. faults	Maximum testing length	Source
TDS1	54	108708	SYS2[16]
TDS2	24	1095.88	SYS4[16]
TDS3	41	4312598	SYS27[16]
TDS4	129	89040	CSR2[1]
TDS5	197	50236822	SYS4[16]
TDS6	136	88682	SYS1[16]
TDS7	104	15369.5	SRC3[1]
TDS8	397	108890	SRC1[1]

Table 4 Software fault-detection group data sets.

Data	No. faults	Testing days	Source
GDS1	54	17	SYS2[16]
GDS2	38	14	SYS3[16]
GDS3	120	19	Release2[28]
GDS4	61	12	Release3[28]
GDS5	9	14	NASA -supported project[27]
GDS6	66	20	DS1[23]
GDS7	58	33	DS2[23]
GDS8	52	30	DS3[23]

4.2 Goodness-of-fit Performances

Goodness-of-fit is a statistical measurement used to assess how well a probability model fits the observed data. It indicates whether the probability model captures the underlying data accurately, while the poor fitting suggests that the model under consideration may not adequately explain the observed data. To investigate the goodness-of-fit for different modeling frameworks, we apply the Akaike information criterion (AIC);

$$\text{AIC} = -2 \ln L(\hat{\theta}) + 2 \times (\text{number of free parameters}), \quad (38)$$

where $\ln L(\hat{\theta})$ is the maximized log-likelihood function. The smaller AIC indicates better goodness-of-fit for the SRM. Figure 2 depicts the mean value functions and the cumulative number of software faults in TDS4. Among the baseline models considered in the finite- and infinite-failure NHPP-, GBP- and GPP-based SRMs, the best SRM in

each category with the minimum AIC was chosen. From this figure, we can see that the modeling framework based on infinite-failure NHPP-based SRMs(orange curve) could not exhibit a good fitting at the beginning of testing, even though it could gradually fit the actual data afterward. On the other hand, the other models could have better fitting performances overall. It is worth noting that the same baseline model shows similar behavior across various modeling frameworks. For example, the baseline model ‘lxvmax’ displays similar patterns in three different finite-failure modeling frameworks. Figure 3 depicts the mean value functions and the cumulative number of software faults in GDS7. It is evident from this figure that the modeling framework based on the infinite-failure NHPP- and GPP-based SRMs (blue and orange curves) could not fit the actual data better than those of other finite- and alternative NHPP-, GBP-, GPP-based SRMs.

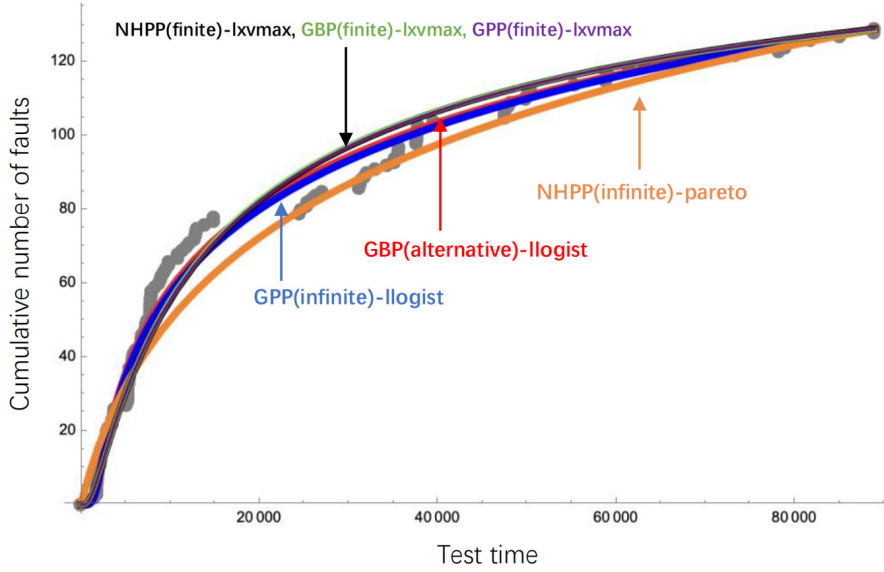


Fig. 2 Behavior of the cumulative number of software faults with the GBP-, GPP- and NHPP-based SRMs in TDS4.

In Table 5, we compare the minimum AIC with 8 time-domain data sets, where the bold numbers mark the best SRM with the smallest AIC in each data set. From the results, the infinite-failure NHPP-based SRMs gave the smallest AIC in 4 cases out of 8 and our alternative GBP-based SRMs showed the smallest AIC in only 3 cases. The infinite-failure GPP-based SRMs did not give the minimum AIC in any case. Thus we conclude that both the infinite-failure NHPP-based SRMs and our alternative GBP-based SRMs are useful to describe the software fault time-domain data. On one hand, we compare the minimum AIC with 8 group data sets in Table 6. From the results, it is seen that our alternative GBP-based SRMs could give the smallest AIC in 5 cases out of 8, and the infinite-failure GPP-based SRMs did the smallest AIC in 2 cases.

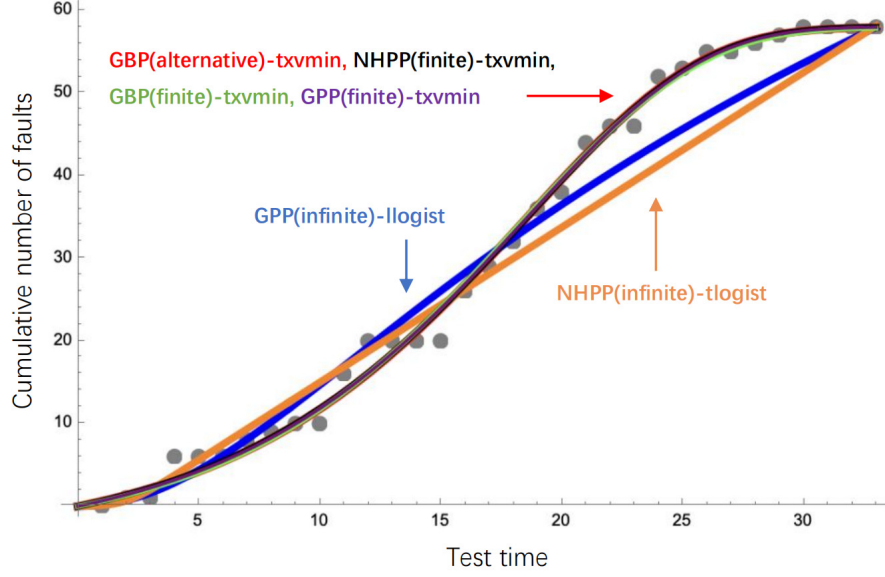


Fig. 3 Behavior of the cumulative number of software faults with the GBP-, GPP- and NHPP-based SRMs in GDS7.

Hence, it can be observed that our alternative GBP-based SRMs could exhibit better fitting performance in many cases.

4.3 Predictive Performance

Predictive performance for SRMs is of paramount importance for evaluating their quantitative software reliability. A highly predictive performing SRM should be able to accurately predict the future data. Evaluating the predictive performance allows us to assess whether the underlying SRM can provide reliable predictions in real-world applications, and enables the effective evaluation of the reliability of the software system. It is worth noting that the SRM with a higher goodness-of-fit to the historical testing data does not always mean that the predictive performance is also higher [13]. Here, we assess the predictive performances using the predictive mean squares error (PMSE), which is defined as the average squared distance between the predicted cumulative number of software faults and the unknown realizations over the prediction length. The PMSE for the time data and group data are defined by

$$\text{PMSE} = \frac{\sqrt{\sum_{i=p+1}^m (E[N(t_i)] - i)^2}}{m - p}, \quad (39)$$

$$\text{PMSE} = \frac{\sqrt{\sum_{i=p+1}^m (E[N(t_i)] - n_i)^2}}{m - p}, \quad (40)$$

Table 5 Goodness-of-fit performance based on (time data).

	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	AIC	Best Model	AIC	Best Model	AIC	Best Model	AIC	Best Model	AIC	Best Model	AIC
TDS1	log	895.305	pareto	895.305	plaw	896.668	lxvmax	898.107	log	897.305	lnorm	898.534
TDS2	plaw	234.967	exp	233.420	tlogist	235.343	txvmin	235.340	plaw	236.967	lnorm	237.110
TDS3	log	1007.100	pareto	1007.095	log	1007.723	exp	1008.200	log	1009.100	llogist	1009.94
TDS4	lxvmax	1822.000	pareto	1838.511	lxvmax	1823.876	llogist	1817.400	lxvmax	1824.000	llogist	1817.410
TDS5	txvmin	5296.620	pareto	5298.808	txvmin	5296.000	txvmin	5295.990	txvmin	5298.620	llogist	5301.650
TDS6	lxvmin	1938.160	pareto	1939.603	plaw	1936.541	gamma	1938.300	lxvmin	1940.160	llogist	1940.650
TDS7	log	1203.523	pareto	1203.523	log	1205.529	lnorm	1206.070	log	1205.093	lnorm	1206.06
TDS8	lxvmax	4720.070	pareto	4750.809	lxvmax	4715.180	llogist	4702.340	lxvmax	4722.070	llogist	4713.700

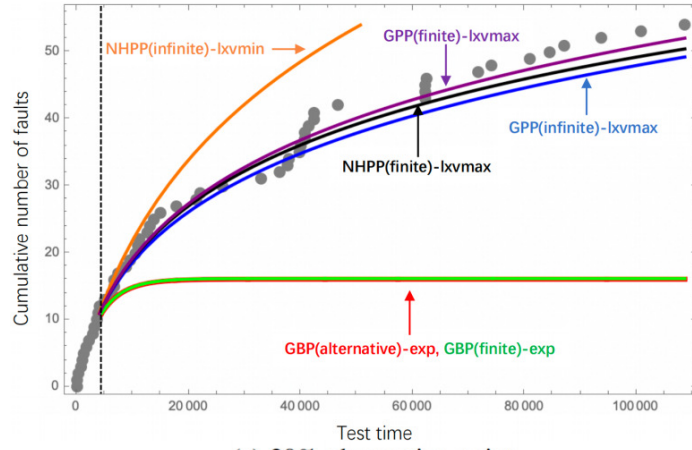
Table 6 Goodness-of-fit performance based on AIC (group data).

	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	AIC	Best Model	AIC	Best Model	AIC	Best Model	AIC	Best Model	AIC	Best Model	AIC
GDS1	llogist	73.053	tlogist	85.339	txvmax	73.187	tnorm	69.923	llogist	75.053	llogist	78.119
GDS2	lxvmax	61.695	llogist	60.674	plaw	62.878	llogist	60.801	lxvmax	63.695	llogist	60.586
GDS3	tlogist	87.275	txvmin	91.919	txvmin	84.881	txvmin	84.877	tlogist	87.277	llogist	99.308
GDS4	tlogist	51.057	txvmin	63.556	plaw	52.576	tlogist	46.767	tlogist	53.057	llogist	66.443
GDS5	exp	29.535	exp	27.753	exp	31.323	tlogist	30.123	exp	31.956	exp	31.953
GDS6	lxvmax	108.831	llogist	107.211	lxvmax	110.815	lxvmax	105.469	lxvmax	110.831	lxvmax	103.876
GDS7	txvmin	123.265	tlogist	138.029	txvmin	124.742	txvmin	119.434	txvmin	125.265	llogist	138.287
GDS8	llogist	117.475	llogist	148.438	llogist	115.963	tlogist	112.357	llogist	119.475	llogist	145.614

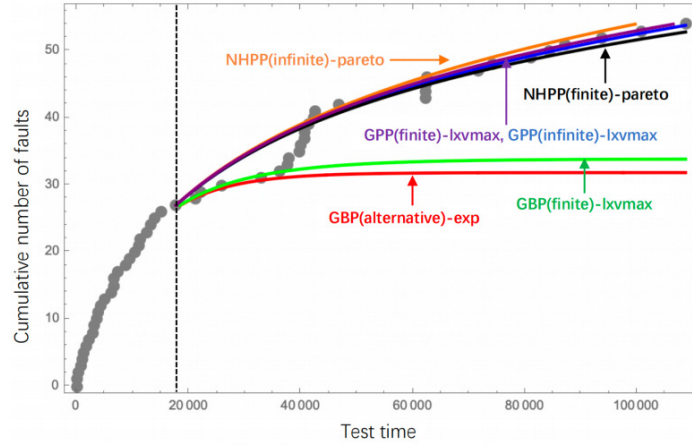
where $p (= 0, 1, \dots, m - 1)$ is the number of observed data and m is the total number of data including the observed (training) data. For the parameter estimation, we still use the maximum likelihood estimation at each observation point t_p . In other words, we use the historical data up to time t_p to estimate the model parameters for SRMs. In our experiments, 20%, 50% and 80% of the software testing time length for the whole data are set as the observation points, so we calculate the PMSE with all the SRMs at each observation point and evaluate the predictive performance of the expected number of software faults in the remaining testing period.

Figure 4 illustrates the predictive behavior of the best SRMs based on the finite-failure NHPP-, GBP-, GPP-based SRMs and our alternative infinite-failure GBP-based SRMs at three different observation points with TDS1. In the 20% observation point, neither the GBP-based SRM nor the infinite-failure NHPP-based SRM showed better predictive performance, while the other three models could show very similar predictive trends. However, in the 50% observation point, both the NHPP-based SRMs and the GPP-based SRMs showed better and very similar predictive trajectories, but the GBP model still flattened out quickly after a short period of growth, leaving it completely off from the actual data. In the 80% observation point, the NHPP- and GPP-based SRMs exhibited almost similar predictive performances, whereas the GBP-based SRMs also demonstrated better predictability. Also, we randomly selected the data set of GDS7 results as an example in the numerical experiment using group data in Figure 5. In the 20% observation point, it is evident that the GBP-based SRM could make inaccurate predictions, as the growing trend curve after the observation point deviated significantly from the actual number of detected faults. In contrast, the infinite-failure GPP-based SRM had more accurate prediction performance than the infinite-failure NHPP-based SRM, and was comparable to the traditional finite-failure SRMs. On the other hand, in the 50% observation point, the prediction behavior of the 6 SRMs exhibited almost similar trends and showed clear upward trends. However, in the 80% observation point, the finite-failure NHPP-, GPP-based SRMs and our alternative infinite-failure GBP-based SRM could demonstrate better predictive performances and gave relatively smooth upward prediction trends, while infinite-failure NHPP- and GPP-based SRMs exhibited significant upward trends. Especially, the traditional NHPP-based SRM, while initially matching the data points, suddenly exhibited a large upward trend thereafter, causing it to deviate completely from the actual data. Moreover, the finite-failure GBP-based SRM appeared as a nearly linear line that has reached a plateau. While it currently did not deviate too far from the actual data, forecasting future failures with this model might prove challenging.

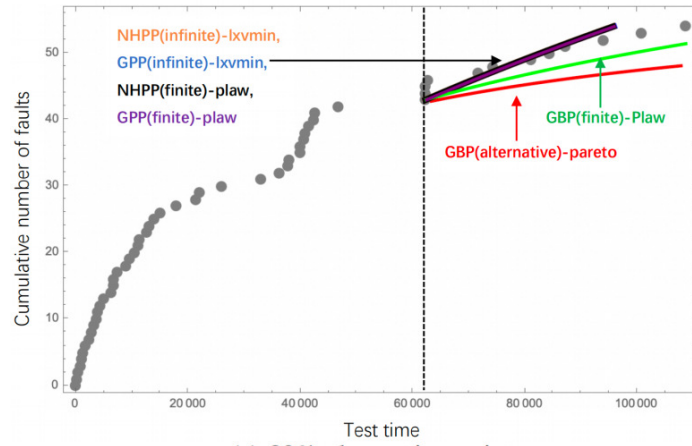
Table 7 compares the predictive performances with 8 software fault time-domain data. We selected the best SRM with the smallest PMSE value in each observation point as the best prediction model. Based on the results presented in Table 7, our alternative NHMP-based SRMs demonstrated smaller PMSE compared to the existing NHPP- and NHMP-based SRMs in 2 out of the 8 data sets in the 20% observation point. However, in the 50% observation point, our NHMP-based SRMs could not give the smallest PMSE in any data set. In the 80% observation point, only our alternative GBP-based SRMs exhibited the smallest PMSE in TDS4. Thus we can see that our



(a) 20% observation point.

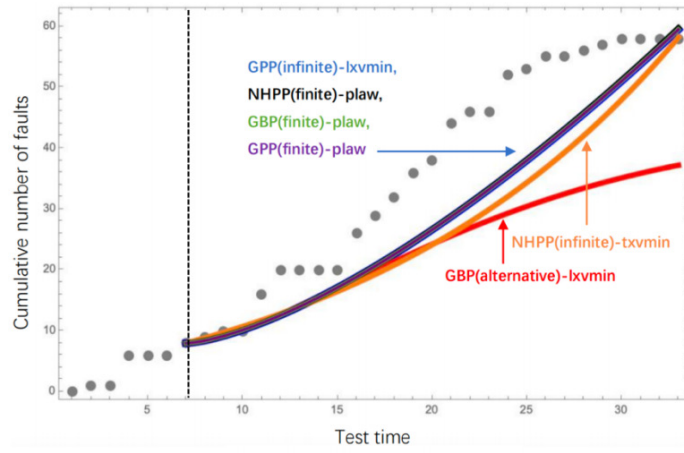


(b) 50% observation point.

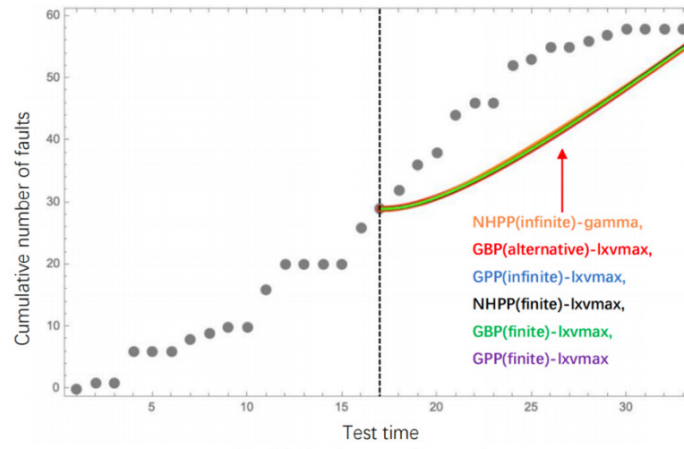


(c) 80% observation point.

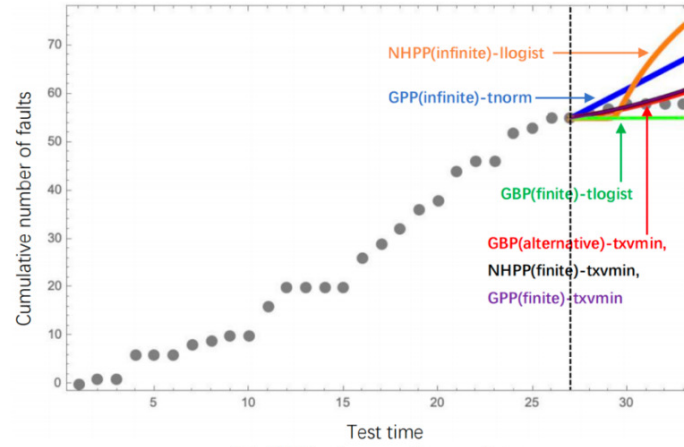
Fig. 4 Predictive behavior of the cumulative number of software faults with infinite-failure NHPP-, GBP- and GPP-based SRMs in TDS1.



(a) 20% observation point.



(b) 50% observation point.



(c) 80% observation point.

Fig. 5 Predictive behavior of the cumulative number of software faults with infinite-failure NHPP-, GBP- and GPP-based SRMs in GDS7.

NHMP-based SRMs proposed in this paper did not outperform in all time-domain data sets. In general, the finite-failure SRMs are suitable for the vast majority of cases.

Then, Table 8 compares the predictive performances with 8 software fault group data. From the results, we can clearly see that our infinite-failure NHMP-based SRMs did not also exhibit the smallest PMSE in the numerical experiments at all observation points. We can say that in almost all cases the smallest PMSE results are obtained with the existing finite-failure SRMs. If one looks at only the results of PMSE, then neither our infinite-failure NHMP-based SRMs nor the infinite-failure NHPP-based SRMs could show satisfactory predictive performances, in spite of the kind of data set, only the finite-failure SRMs could exhibit better predictive performances.

However, it should be emphasized that the predictive performances in Tables 7 and 8 are based on the minimum PMSE. It is important to note that choosing the best SRM with the minimum PMSE is impossible because one cannot know the minimum PMSE at each observation point. Then it is common to apply the best goodness-of-fit model at each observation point and to use it for the prediction. Hence, we assume We assume the model selection with the minimum AIC at each observation point and compare the resulting PMSE in the remaining prediction period.

Tables 9 and 10 present the predictive performances when the baseline model was selected with the minimum AIC at each observation point with time-domain data. From the results in Table 7 compared a total of 24 cases (8 time-domain data sets $\times$ 3 kinds of models), we find that the existing finite-failure SRMs and infinite-failure NHPP-based SRMs could provide the minimum AIC in 18 cases. However, focusing on the PMSE by selecting unknown best prediction models with the minimum AIC, the existing finite-failure SRMs and infinite-failure NHPP-based SRMs gave the best results in only 12 cases. In other words, our infinite-failure NHMP-based SRMs can exhibit the smallest PMSE in half of the cases under the realistic model selection scenario. In particular, in many cases, our infinite-failure NHMP-based SRMs gave significantly better prediction results than other SRMs. For example, check the results for the infinite-failure GPP-based SRMs in TDS4 with 20% observation point and the infinite-failure GBP-based SRMs in TDS6 with 50% observation point, etc. in Tables 9 and 10. Moreover, we can reconfirm that the SRMs with the smallest AIC do not necessarily give the better predictive performances, e.g., the infinite-failure NHPP-based SRMs in TDS1~TDS6 with 50% observation point exhibited the best predictive performances only in TDS5, although they all gave the smallest AIC in 6 cases out of 8.

Similarly, we also compared the predictive performances when the model selection based on the minimum AIC is applied for the group data in Tables 11 and 12. From the results in Tables 11 and 12, the infinite-failure GPP-based SRMs could provide the minimum PMSE in 2 cases out of 8 and the NHPP-based SRMs provided the minimum PMSE in 4 cases, so the predictive performance of the infinite-failure SRMs is competitive in most cases with 20% observation point. For the results with 50% observation point, our infinite-failure NHMP-based SRMs gave the best results in 4 cases, whereas the infinite-failure GPP-based SRMs exhibited the smallest PMSE in 3 cases. In addition, the infinite-failure GPP-based SRMs provided larger PMSE than the infinite-failure NHPP-based SRMs in only GDS7, while the PMSE value in

Table 7 Predictive performance based on PMSE (time data).

20% Observation Point												
	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE
TDS1	lxvmax	0.323	exp	1.613	exp	3.051	exp	3.131	lxvmax	0.252	lxvmax	0.446
TDS2	log	0.743	pareto	0.781	pareto	0.729	pareto	0.778	log	0.742	exp	0.787
TDS3	lxvmax	1.371	llogist	2.508	log	1.574	pareto	2.106	lxvmax	1.592	lxvmax	1.581
TDS4	lxvmax	1.973	llogist	3.573	lxvmax	2.778	llogist	5.398	lxvmax	1.973	llogist	0.560
TDS5	llogist	0.389	exp	0.409	tnorm	0.354	pareto	0.434	llogist	0.385	pareto	0.414
TDS6	lxvmax	1.361	pareto	2.895	lxvmax	1.640	lxvmax	1.690	lxvmax	1.978	lxvmax	1.770
TDS7	lnorm	6.270	llogist	3.937	pareto	0.618	lxvmax	1.146	lxvmax	5.495	lxvmax	5.327
TDS8	llogist	7.756	llogist	8.508	lxvmax	8.995	lxvmax	9.049	llogist	7.756	llogist	1.200
50% Observation Point												
	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE
TDS1	pareto	0.406	pareto	0.488	lxvmax	1.984	exp	2.322	pareto	0.428	llogist	0.416
TDS2	txvmin	0.930	txvmax	0.804	tnorm	1.085	tnorm	1.088	txvmin	0.933	tlogist	1.008
TDS3	log	13.937	lxvmax	0.876	txvmin	5.167	tlogist	4.087	exp	15.115	gamma	34.955
TDS4	gtlogist	1.581	llogist	1.393	lnorm	1.231	tlogist	2.465	llogist	1.338	lxvmax	6.360
TDS5	log	1.743	pareto	1.773	log	1.131	exp	1.728	exp	2.153	pareto	1.755
TDS6	log	0.383	pareto	0.389	log	0.389	gamma	0.617	pareto	0.359	pareto	2.410
TDS7	log	0.350	pareto	0.357	pareto	3.183	exp	3.274	log	0.349	pareto	0.357
TDS8	tnorm	5.764	llogist	4.575	tlogist	2.877	txvmax	18.304	tnorm	5.729	llogist	26.676
80% Observation Point												
	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE
TDS1	plaw	0.511	lxvmin	0.559	lxvmax	0.667	pareto	1.206	plaw	0.511	lxvmin	0.555
TDS2	log	1.239	llogist	0.303	log	1.234	pareto	1.485	exp	1.243	pareto	1.491
TDS3	plaw	0.621	lxvmin	0.690	tlogist	0.807	llogist	1.582	plaw	0.621	lxvmin	0.693
TDS4	lxvmax	0.655	pareto	1.784	log	0.352	lxvmax	0.309	lxvmax	0.655	llogist	0.608
TDS5	log	0.682	pareto	0.699	pareto	0.684	lxvmax	0.706	log	0.681	lnorm	0.699
TDS6	lxvmax	0.563	lxvmin	0.845	log	1.028	pareto	1.064	lxvmax	0.560	lxvmax	0.578
TDS7	lxvmax	0.529	pareto	0.562	pareto	0.567	lxvmax	0.597	lxvmax	0.529	pareto	0.563
TDS8	lxvmax	0.401	pareto	2.657	txvmax	1.741	llogist	2.199	lxvmax	0.401	llogist	1.757

Table 8 Predictive performance based on PMSE (group data).

20% Observation Point												
	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE
GDS1	plaw	3.689	lxvmin	3.953	gamma	3.686	gamma	7.846	plaw	3.572	gamma	3.943
GDS2	lxvmax	1.441	llogist	1.987	lxvmax	1.595	tlogist	3.554	gamma	1.728	pareto	9.141
GDS3	tlogist	3.343	gamma	3.276	tlogist	3.177	lnorm	11.893	tlogist	2.754	tlogist	4.682
GDS4	exp	3.436	tlogist	2.934	log	3.436	gamma	7.8641	exp	3.210	txvmin	3.775
GDS5	pareto	0.432	pareto	0.468	log	0.432	pareto	0.526	exp	0.427	exp	0.467
GDS6	tlogist	2.340	llogist	1.265	log	1.847	tlogist	1.902	lxvmax	4.487	llogist	1.417
GDS7	plaw	0.957	txvmin	1.584	plaw	3.871	lxvmin	3.357	plaw	0.945	lxvmin	0.980
GDS8	txvmin	4.032	txvmin	4.200	tlogist	3.532	lxvmin	3.877	tnorm	3.687	pareto	6.958
50% Observation Point												
	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE
GDS1	tnorm	0.893	pareto	4.210	lxvmin	0.921	exp	0.948	tnorm	0.891	tlogist	11.522
GDS2	plaw	0.638	lxvmin	0.729	tlogist	2.047	txvmax	2.901	plaw	0.635	lnorm	0.652
GDS3	lxvmax	6.832	gamma	9.218	lxvmax	6.100	lxvmin	10.048	log	10.777	lxvmax	8.924
GDS4	exp	3.522	pareto	4.108	exp	3.443	pareto	4.101	log	4.207	pareto	4.103
GDS5	exp	0.194	pareto	0.223	pareto	0.193	txvmin	0.211	pareto	0.193	pareto	0.221
GDS6	log	2.047	gamma	1.372	log	5.719	gamma	6.010	plaw	2.223	tnorm	1.675
GDS7	lxvmax	1.096	gamma	1.261	lxvmax	1.117	lxvmax	1.212	lxvmax	1.081	lxvmax	1.146
GDS8	txvmin	1.131	llogist	3.228	lxvmin	1.022	lxvmin	1.098	txvmin	1.133	tlogist	27.725
80% Observation Point												
	NHPP (finite)		NHPP (infinite)		GBP (finite)		GBP (alternative)		GPP (finite)		GPP (infinite)	
	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE	Best Model	PMSE
GDS1	lnorm	0.531	llogist	1.478	plaw	1.071	tlogist	1.816	lnorm	0.531	llogist	1.728
GDS2	exp	0.295	txvmin	0.393	tlogist	0.295	pareto	0.587	tlogist	0.294	lnorm	0.400
GDS3	tnorm	0.230	txvmin	3.210	gtlogist	0.645	txvmax	1.635	tnorm	0.228	llogist	4.084
GDS4	tnorm	0.589	llogist	2.516	tlogist	0.781	txvmax	1.537	tnorm	0.589	tlogist	6.097
GDS5	plaw	0.169	lxvmax	0.173	llogist	0.169	exp	0.884	gtlogist	0.169	lxvmin	0.227
GDS6	txvmax	0.741	llogist	1.239	lxvmax	0.943	lxvmax	0.641	lxvmax	0.945	lxvmax	0.785
GDS7	txvmin	0.818	llogist	1.572	tlogist	0.143	txvmin	0.166	txvmin	0.817	tnorm	2.614
GDS8	lxvmax	0.325	llogist	1.027	lxvmax	0.325	tlogist	1.289	lxvmax	0.314	lxvmax	1.757

Table 9 Predictive performance based on the baseline model with minimum AIC (time data).

20% Observation Point												
	NHPP (finite)			GBP (finite)			GBP (alternative)			GPP (finite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE
TDS1	log	141.591	1.577	plaw	142.559	3.599	exp	142.999	3.131	log	143.591	1.575
TDS2	plaw	35.327	2554.649	plaw	37.055	1.408	tlogist	35.132	2.522	plaw	37.327	1.091e+10
TDS3	plaw	187.583	4.817	plaw	189.398	1.769	exp	189.469	1.662	plaw	189.583	4.791
TDS4	llogist	291.964	5.375	txvmax	292.373	5.624	tlogist	290.85	5.502	llogist	293.965	5.375
TDS5	plaw	1046.910	0.464	plaw	1048.895	0.427	exp	1048.920	0.572	plaw	1048.910	0.470
TDS6	plaw	311.745	3.573	plaw	313.358	4.116	exp	314.461	5.490	plaw	313.745	3.576
TDS7	plaw	191.169	11.723	plaw	193.027	2.485	exp	193.046	1.422	plaw	193.169	11.805
TDS8	txvmax	752.937	9.350	txvmax	753.232	9.784	tlogist	752.913	9.299	txvmax	754.937	9.350
50% Observation Point												
	NHPP (finite)			GBP (finite)			GBP (alternative)			GPP (finite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE
TDS1	log	403.240	0.471	plaw	403.655	2.369	exp	403.69	2.322	log	405.240	0.471
TDS2	plaw	123.241	0.990	plaw	124.923	1.366	tlogist	122.522	1.645	plaw	125.241	0.990
TDS3	plaw	445.124	33.395	tlogist	447.172	5.389	tlogist	447.147	4.087	plaw	446.531	35801.400
TDS4	lxvmax	764.771	4.111	plaw	766.231	3.311	tlogist	763.999	2.465	lxvmax	766.771	4.113
TDS5	plaw	2601.980	2.834	plaw	2604.140	2.418	tlogist	2604.910	2.925	plaw	2603.510	5.771
TDS6	plaw	859.945	3.884	plaw	861.832	2.052	gamma	863.515	0.617	plaw	861.945	3.886
TDS7	exp	527.080	2.993	plaw	525.815	3.800	txvmin	525.396	3.954	exp	529.068	3.069
TDS8	plaw	1900.550	89.228	plaw	1901.390	9.748	lnorm	1903.020	25.030	plaw	1902.550	89.200
80% Observation Point												
	NHPP (finite)			GBP (finite)			GBP (alternative)			GPP (finite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE
TDS1	log	690.114	0.702	plaw	690.889	1.370	exp	692.286	1.783	log	692.114	0.702
TDS2	tlogist	176.266	8.489	tlogist	178.264	8.607	tlogist	178.263	10.341	txvmin	177.889	14.386
TDS3	log	769.639	0.829	tlogist	768.020	0.807	exp	768.956	1.746	log	771.641	0.831
TDS4	lxvmax	1372.520	0.655	lxvmax	1374.290	0.751	llogist	1369.850	0.442	lxvmax	1374.520	0.655
TDS5	exp	4203.050	0.683	plaw	4204.420	0.698	exp	4204.860	0.723	exp	4205.050	0.683
TDS6	llogist	1478.500	0.943	plaw	1477.230	1.343	gamma	1479.090	1.585	tlogist	1479.313	2.368
TDS7	log	920.101	0.536	log	922.031	0.608	txvmin	923.409	0.597	log	922.101	0.536
TDS8	llogist	3465.000	3.212	txvmax	3449.830	1.741	lnorm	3459.970	2.199	llogist	3467.000	3.212

Table 10 Continuation of Table 9 (time data).

20% Observation Point						
	NHPP (infinite)			GPP (infinite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE
TDS1	exp	139.683	15.223	exp	143.683	15.249
TDS2	lxvmax	33.490	1.399	lxvmax	38.427	2.929
TDS3	exp	185.583	4.915	exp	189.584	4.973
TDS4	tlogist	293.584	22.285	llogist	293.565	0.560
TDS5	exp	1044.923	0.409	txvmax	1050.86	1.694
TDS6	lxvmin	311.745	3.605	llogist	315.746	2.615
TDS7	exp	189.228	14.696	exp	193.229	14.721
TDS8	lxvmin	758.536	119.663	tlogist	757.556	51.823
50% Observation Point						
	NHPP (infinite)			GPP (infinite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE
TDS1	pareto	403.239	0.488	lxvmax	406.848	0.740
TDS2	lxvmax	118.597	1.485	lxvmax	122.246	1.463
TDS3	exp	444.960	14.569	exp	446.610	25954.800
TDS4	tlogist	763.495	29.209	llogist	766.054	11.022
TDS5	exp	2600.905	1.781	txvmax	2603.250	2.132
TDS6	lxvmin	859.945	3.942	gamma	863.942	4.126
TDS7	pareto	528.954	0.357	llogist	532.569	0.471
TDS8	lxvmin	1900.555	89.679	tnorm	1895.860	104.490
80% Observation Point						
	NHPP (infinite)			GPP (infinite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE
TDS1	exp	690.113	0.764	lnorm	693.644	0.616
TDS2	txvmin	174.262	10.341	tlogist	177.808	21.179
TDS3	pareto	769.639	0.921	llogist	772.233	1.008
TDS4	pareto	1388.266	1.784	llogist	1369.750	0.608
TDS5	exp	4201.458	0.773	exp	4205.460	0.773
TDS6	pareto	1479.293	1.056	llogist	1480.680	0.770
TDS7	pareto	920.101	0.562	lxvmax	923.763	0.572
TDS8	tlogist	3504.080	2.657	llogist	3473.100	1.437

GDS5 was same as that of infinite-failure NHPP-based SRMs. Except in the above case, the predictive performances of infinite-failure GPP-based SRMs were superior to those of the infinite-failure NHPP-based SRMs in 6 data sets. When the observation point is set to 80%, the results in all cases showed the predictive performances of the finite-failure SRMs to be better, although the infinite-failure GPP-based SRMs outperformed the infinite-failure NHPP-based SRMs in 6 data sets. Furthermore, the NHMP-based SRMs (GBP and GPP) demonstrated significantly better predictive performances than the NHPP-based SRMs in several cases, in GDS8 with 20% observation point, GDS1 and GDS8 with 50% observation point, and GDS4 and GDS7 with 80% observation point. Combining the results in Tables 7 and 8, we can conclude that our infinite-failure NHMP-based SRMs did not always outperform the other existing SRMs in the case where the best baseline model at each observation point is unknown in advance. However, from the results in Tables 9-12, the performance of our infinite-failure NHMP-based SRMs could exhibit sufficient competitiveness in almost half of the cases where the best baseline model at each observation point was known in advance.

Table 11 Predictive performance based on the baseline model with minimum AIC (group data).

20% Observation Point												
	NHPP (finite)			GBP (finite)			GBP (alternative)			GPP (finite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE
GDS1	plaw	10.807	3.689	plaw	12.637	6.167	exp	12.992	8.492	plaw	12.807	3.572
GDS2	plaw	10.864	2.145	plaw	12.378	2.219	llogist	13.293	7.876	plaw	12.865	327.513
GDS3	plaw	17.199	6.764	plaw	19.031	12.026	exp	19.124	14.751	plaw	19.199	6.758
GDS4	plaw	10.649	8.017	exp	10.629	11.147	exp	11.350	11.419	plaw	12.649	7.981
GDS5	plaw	6.000	2.415e+11	plaw	7.976	7.979	exp	9.078	5.916	plaw	8.000	3.606e+6325m
GDS6	lnorm	20.660	34.922	tlogist	21.781	5.540	lxvmax	16.277	5.916	lnorm	22.660	5.146
GDS7	txvmin	16.959	6.600	txvmin	18.953	6.599	llogist	24.914	6.186	txvmin	18.959	6.601
GDS8	plaw	6.709	2.081e+13	plaw	8.696	3.369	tlogist	10.404	6.434	plaw	8.730	2.825e+4245m
50% Observation Point												
	NHPP (finite)			GBP (finite)			GBP (alternative)			GPP (finite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE
GDS1	tlogist	34.645	1.437	txvmin	33.667	3.531	txvmin	33.665	3.925	tlogist	36.649	1.437
GDS2	lxvmax	31.051	2.633	plaw	29.471	3.455	tlogist	29.545	3.647	lxvmax	33.048	2.633
GDS3	gtlogist	49.161	20.583	gtlogist	51.099	18.316	tlogist	51.066	10.214	plaw	49.107	20.559
GDS4	tlogist	30.560	22.973	plaw	33.087	7.684	txvmin	32.474	12.562	plaw	31.685	20.962
GDS5	plaw	16.878	0.796	plaw	19.911	0.732	lxvmin	18.406	1.015	plaw	18.882	0.770
GDS6	lxvmax	40.521	5.643	lxvmax	40.768	5.813	llogist	38.982	6.176	lxvmax	42.521	5.642
GDS7	plaw	70.521	1.790	plaw	72.332	1.051	lxvmin	74.286	1.934	plaw	72.521	1.768
GDS8	txvmin	65.713	1.131	gtlogist	60.151	11.712	tlogist	64.323	1.766	txvmin	67.127	1.133
80% Observation Point												
	NHPP (finite)			GBP (finite)			GBP (alternative)			GPP (finite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE	Best Model	AIC	PMSE
GDS1	tlogist	55.069	1.406	txvmax	53.232	1.651	tlogist	52.429	1.816	tlogist	57.069	18.845
GDS2	lxvmax	52.523	0.417	plaw	54.251	0.655	llogist	51.482	0.658	lxvmax	54.526	0.419
GDS3	txvmin	75.292	0.887	tlogist	73.881	1.485	txvmin	73.878	1.844	txvmin	77.293	0.887
GDS4	txvmin	42.548	0.828	tnorm	40.788	1.242	txvmin	40.241	1.812	txvmin	44.548	0.827
GDS5	exp	24.272	0.286	exp	25.495	0.664	llogist	24.282	0.935	gtlogist	26.272	0.169
GDS6	lxvmax	96.179	0.945	lxvmax	98.179	0.943	llogist	93.345	0.654	lxvmax	98.172	0.945
GDS7	plaw	111.458	4.137	tlogist	112.304	0.143	txvmin	112.301	0.166	plaw	113.458	4.137
GDS8	tlogist	100.326	0.855	tlogist	99.291	1.022	lnorm	92.465	1.389	tlogist	102.326	0.855

Table 12 Continuation of Table 11 (group data).

20% Observation Point						
	NHPP (infinite)			GPP (infinite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE
GDS1	exp	9.085	6.686	exp	13.085	6.683
GDS2	tlogist	10.865	18.9392	lxvmax	14.865	16.627
GDS3	gamma	16.651	3.276	llogist	21.200	7.202
GDS4	exp	9.669	3.778	exp	13.669	3.778
GDS5	exp	5.386	0.468	exp	9.386	0.467
GDS6	llogist	27.055	1.265	llogist	24.664	1.418
GDS7	tlogist	23.849	2.093	llogist	26.369	5.553
GDS8	tlogist	6.614	56.2505	lxvmax	10.614	7.542
50% Observation Point						
	NHPP (infinite)			GPP (infinite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE
GDS1	lxvmin	35.089	21.134	tlogist	37.495	11.523
GDS2	lxvmax	31.770	2.156	llogist	32.530	1.741
GDS3	exp	47.312	10.249	exp	50.826	15.545
GDS4	txvmin	28.559	26.944	exp	31.741	23.323
GDS5	exp	15.365	0.223	exp	19.365	0.223
GDS6	llogist	48.831	4.929	llogist	45.976	4.413
GDS7	gamma	70.237	1.261	tlogist	74.001	2.467
GDS8	lxvmin	68.490	74.881	tlogist	71.686	27.725
80% Observation Point						
	NHPP (infinite)			GPP (infinite)		
	Best Model	AIC	PMSE	Best Model	AIC	PMSE
GDS1	lxvmin	69.767	4.397	llogist	64.071	1.728
GDS2	llogist	51.400	1.171	llogist	51.371	0.496
GDS3	lxvmin	79.123	5.374	llogist	82.066	4.084
GDS4	exp	44.859	6.121	exp	48.858	6.124
GDS5	lxvmax	22.768	0.679	exp	26.328	0.273
GDS6	llogist	95.606	1.239	llogist	91.289	1.300
GDS7	lxvmin	111.458	4.728	lnorm	115.219	4.204
GDS8	tlogist	133.367	3.472	llogist	128.143	1.877

5 Conclusions

In this paper, we have proposed novel SRMs based on NHMPs, which were categorized into GBPs and GPPs. These models have been developed under the assumption that the baseline intensity functions were given by the failure rate functions of software failure time distribution. We have compared our infinite-failure NHMP-based SRMs with the existing NHPP- and NHMP-based SRMs in terms of goodness-of-fit and predictive performances. We have compared our infinite-failure NHMP-based SRMs under 11 baseline software failure time distributions in a comprehensive experiment with actual software development project data, including both time-domain data and group data.

Our experimental results have suggested that the NHMP-based SRMs developed in this paper consistently demonstrated better goodness-of-fit in almost all cases. In the predictive performance of these SRMs, although our infinite-failure NHMP-based SRMs could exhibit sufficient competitiveness in almost half of the cases, overall all the infinite-failure SRMs did not outperform the finite-failure SRMs. However, if one focuses on the performance comparison with infinite-failure SRMs, our infinite-failure GPP-based SRMs have outperformed the infinite-failure NHPP-based SRMs in most

cases. Therefore the infinite-failure NHMP-based SRM can also be a good choice when an infinite-failure model is required.

Based on our analysis, we have concluded that the NHMP-based SRMs provided better predictive performance than the NHPP-based SRMs in most cases. Thus, it would be worthwhile to explore other NHMP-based SRMs besides GBP and GPP. In our modeling framework, we have implicitly assumed the linear transition rates in GBPs and GPPs. In the future direction, we will consider the different NHMP-based SRMs with nonlinear transition rates do not belong to GBP and GPP anymore. The excellent performances of the generalized NHMP-based SRMs will open up new possibilities for their applications in the field of software reliability engineering, so we expect further advancements in the field of software trustworthiness in the future by exploring the widely acceptable SRMs.

Declarations

- Funding
No funding was obtained for this study.
- Competing interests
The authors have no competing interests to declare that are relevant to the content of this article.
- Ethics approval
Not applicable.
- Consent to participate
Not applicable.
- Consent for publication
Not applicable.
- Availability of data and materials
The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.
- Code availability
Not applicable.
- Authors' contributions
Conceptualization: Nanxiang Qiu, Siqiao Li, Tadashi Dohi and Hiroyuki Okamura.; Methodology: Nanxiang Qiu, Siqiao Li, Tadashi Dohi and Hiroyuki Okamura.; Formal analysis and investigation: Nanxiang Qiu, Siqiao Li, Tadashi Dohi and Hiroyuki Okamura.; Writing-original draft preparation: Nanxiang Qiu.; Writing-review and editing: Tadashi Dohi. All authors have read and agreed to the published version of the manuscript.

References

- [1] M. R. Lyu (ed.). (1996). *Handbook of Software Reliability Engineering*, McGraw-Hill, New York.
- [2] J. D. Musa, A. Iannino, and K. Okumoto. (1987). *Software Reliability, Measurement, Prediction, Application*, McGraw-Hill, New York.

- [3] M. Xie. (1991). *Software Reliability Modeling*, World Scientific, Singapore.
- [4] A. A. Abdel-Ghaly, P. Y. Chan, and B. Littlewood. (1986). Evaluation of competing software reliability predictions, *IEEE Transactions on Software Engineering*, vol. SE-12, no. 9, pp. 950–967.
- [5] J. A. Achcar, D. K. Dey, and M. Niverthi. (1998). A Bayesian approach using non-homogeneous Poisson processes for software reliability models, in *Frontiers in Reliability*, A. P. Basu, K. S. Basu, and S. Mukhopadhyay (eds.), pp. 1–18, World Scientific, Singapore.
- [6] P. Erto, M. Giorgio, and A. Lepore. (2020). The generalized inflection S shaped software reliability growth model, *IEEE Transactions on Reliability*, vol. 69, no. 1, pp. 228–244.
- [7] Y. L. Gat. (2016). *Recurrent Event Modeling based on the Yule Process Application to Water Network Asset Management*, vol. 2, John Wiley & Sons, London.
- [8] A. L. Goel, and K. Okumoto. (1979). Time-dependent error-detection rate model for software reliability and other performance measures, *IEEE Transactions on Reliability*, vol. R-28, no. 3, pp. 206–211.
- [9] A. L. Goel. (1985). Software reliability models: assumptions, limitations and applicability, *IEEE Transactions on Software Engineering*, vol. SE-11, no. 12, pp. 1411–1423.
- [10] S. S. Gokhale, and K. S. Trivedi. (1998). Log-logistic software reliability growth model, in *Proceedings of the 3rd IEEE International Symposium on High-Assurance Systems Engineering (HASE-1998)*, pp. 34–41, IEEE CPS.
- [11] D. G. Kendall. (1948). On the generalized birth-and-death process, *Annals of Mathematical Statistics*, vol. 19, pp. 1–5.
- [12] H. Konno. (2010). On the exact solution of a generalized Polya process, *Advances in Mathematical Physics*, vol. 2010, Article ID 504267, 12 pages.
- [13] S. Li, T. Dohi and H. Okamura. (2023). Are infinite-failure NHPP-based software reliability models useful?, *Software*, vol. 2, no. 1, 1–18.
- [14] S. Li, T. Dohi and H. Okamura. (2022). Non-homogeneous Markov process modeling for software reliability assessment, *IEEE Transactions on Reliability*, in press.
- [15] B. Littlewood. (1984). Rationale for a modified Duane model, *IEEE Transactions on Reliability*, vol. R-33, no. 2, pp. 157–159.
- [16] J. D. Musa. (1979). Software Reliability Data, *Technical Report in Rome Air Development Center*.

- [17] J. D. Musa, and K. Okumoto. (1984). A logarithmic Poisson execution time model for software reliability measurement, in *Proceedings of the 7th International Conference on Software Engineering (ICSE-1984)*, pp. 230–238, ACM.
- [18] M. Ohba. (1984). Inflection S-shaped software reliability growth model, in *Stochastic Models in Reliability Theory*, S. Osaki and Y. Hatoyama (eds.), pp. 144–165, Springer-Verlag, Berlin.
- [19] K. Ohishi, H. Okamura, and T. Dohi. (2009). Gompertz software reliability model: estimation algorithm and empirical validation, *Journal of Systems and Software*, vol. 82, no. 3, pp. 535–543.
- [20] H. Okamura and T. Dohi. (2008). Software reliability modeling based on mixed Poisson distributions, *International Journal of Reliability, Quality and Safety Engineering*, vol. 15, no. 1, pp. 9–32.
- [21] H. Okamura, T. Dohi, and S. Osaki. (2013). Software reliability growth models with normal failure time distributions, *Reliability Engineering and System Safety*, vol. 116, pp. 135–141.
- [22] H. Okamura and T. Dohi. (2013). SRATS: Software reliability assessment tool on spreadsheet, in *Proceedings of the 2013 IEEE 24th International Symposium on Software Reliability Engineering (ISSRE-2013)*, pp. 100–107, IEEE CPS.
- [23] H. Okamura, Y. Etani, and T. Dohi. (2011). Quantifying the effectiveness of testing efforts on software fault detection with a logit software reliability growth model, in *Proceedings of 2011 Joint Conference of the 21st International Workshop on Software Measurement (IWSM 2011) and the 6th International Conference on Software Process and Product Measurement (MENSURA-2011)*, pp. 62–68, IEEE CPS.
- [24] N. Qiu, S. Li, T. Dohi and H. Okamura. (2023). Another look at non-homogeneous Markovian software reliability modeling, in *Proceedings of The 10th International Conference on Dependable Systems and Their Applications (DSA 2023)*, IEEE CPS.
- [25] J. G. Shanthikumar. (1981). A general software reliability model for performance prediction, *Microelectronics and Reliability*, vol. 21, no. 5, pp. 671–682.
- [26] J. G. Shanthikumar. (1983). Software reliability models: A review, *Microelectronics and Reliability*, vol. 23, no. 5, pp. 903–943.
- [27] M. A. Vouk. (1992). Using reliability models during testing with non-operational profile, in *Proceedings of the 2nd Bell-core/Purdue Workshop on Issues in Software Reliability Estimation*, pp. 254–266.

- [28] A. Wood. (1996). Predicting software reliability, *IEEE Computer*, vol. 20, no. 11, pp. 69–77.
- [29] S. Yamada, M. Ohba, and S. Osaki. (1983). S-shaped reliability growth modeling for software error detection, *IEEE Transactions on Reliability*, vol. R-32, no. 5, pp. 475–478.
- [30] M. Zhao, and M. Xie. (1996). On maximum likelihood estimation for a general non-homogeneous Poisson process, *Scandinavian Journal of Statistics*, 1996, vol. 23, no. 4, pp. 597–607 .